

SAMPLING THEORY

A **population** consists of counting all the members (for example, IQ scores of all students in YTU, the weight of all people in a city) in which we are interested. A measure such as mean, standard deviation, a type of proportion, etc., that summarizes the population is called a **population parameter** or simply a **parameter**.

We can extract information about a population

1. By **full enumeration** if possible (if the population has a small number of members, it has no high cost according to time or money, and the measurement of data does not damage the products we are interested in).
2. By choosing a **sample** from it. This process is called **sampling**. A **sample** is a subset of the population, and it should be representative of the population. The measures (mean, standard deviation, etc.) of a sample are called the **sample statistic** or simply **statistic**.

There are two types of sampling methods:

1. **Non-Stochastic Methods**
 - a. **Convenience Sampling**: Members of the population are chosen based on their relative ease of access (Ex: shoppers at a single mall)
 - b. **Judgmental Sampling**: Choosing a sample from a population according to personal experiences and thoughts (Ex: choosing university students for market research of a new product)
 - c. **Quota Sampling**: A quota is determined for the sampling (Ex: A quota of 50% women)
 - d. **Snowball Sampling**: A member is selected randomly. This member refers to a contact. Then this contact refers to another contact and so on (Ex: Choosing the people using the drug)
2. **Stochastic Methods**:
 - a. **Simple Random Sampling**: Each member is chosen randomly. Here all possible samples selected from a population are equally likely.
 - b. **Stratified Sampling**: If the population is not homogenous, the population is divided into subgroups (stratifies) according to their inhomogeneity, and then members are randomly selected from each group.
 - c. **Systematic Sampling**: The sampling starts by randomly selecting a member from the list of population and then next k. member in the list is selected and so on. Here k is the sampling interval.
 - d. **Cluster Sampling**: The population is divided into clusters, then members are randomly selected from each cluster. Here, clustering is not made according to their inhomogeneity. (Ex: If the population is spread over a large geographical area)

SAMPLING DISTRIBUTION

Let a sample of size n be randomly selected from a population of size N and let $\hat{\theta}$ be a statistic of the sample. Consider all possible combinations of the samples that can be selected from this population. If their statistics are

$$\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3, \dots, \hat{\theta}_{\binom{N}{n}},$$

then this distribution is called **the sampling distribution of the statistic $\hat{\theta}$** .

1. Sampling Distribution of the Sample Mean

When all possible samples of size n are selected from a population of size N , and the mean ($\bar{X}_i$) of each sample is calculated, we obtain the **sampling distribution of the sample means**. In this case, the statistic is

$$\hat{\theta} = \bar{X} = \frac{1}{n} \sum_{j=1}^n X_j$$

where $X_j, j = 1, 2, \dots, n$ are the members of the chosen sample, and $\bar{X}$ is the random variable of the sampling distribution of the sample means. If the samples are selected without replacing them back, the sampling distribution is

$$\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_{\binom{N}{n}}.$$

If the samples are selected with replacing them back, then the sampling distribution is

$$\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_{N^n}.$$

Theorem: If the samples of size n are selected without replacement in a population of size N , the mean and standard deviation of $\bar{X}$ are

$$E(\bar{X}) = \mu, \quad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}},$$

where μ is the mean of the population, σ is the standard deviation of the population, $\sigma_{\bar{X}}$ is the standard deviation of the sampling distribution of the sample means. If the samples are selected with replacement, then:

$$E(\bar{X}) = \mu, \quad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}.$$

The term

$$\sqrt{\frac{N-n}{N-1}}$$

is called the finite population correction factor. Note that,

$$\lim_{N \rightarrow \infty} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{\sigma}{\sqrt{n}}$$

If $n \leq 0.05 N$, we use the formulas for the samples that be replaced back, although the samples are selected without replacing them back. From now on, we assume that samples are selected with replacements.

The proof for the mean and standard deviation of the sampling distribution $\bar{X}$ for the samples that are selected with replacements:

$$E(\bar{X}) = E\left\{\frac{1}{n} \sum_{i=1}^n X_i\right\} = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} n\mu = \mu$$

$$Var(\bar{X}) = \sigma_{\bar{x}}^2 = Var\left\{\frac{1}{n}\sum_{i=1}^n X_i\right\} = \frac{1}{n^2}\sum_{i=1}^n Var(X_i) = \frac{1}{n^2}\sum_{i=1}^n \sigma^2 = \frac{1}{n^2}n\sigma^2 = \frac{\sigma^2}{n} \Rightarrow \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Example: Let the samples with size of 3 be selected without replacing them back into a population consisting of the numbers 2, 4, 10, 12, 18. Create the sampling distribution of the sample mean and then calculate the mean and the standard deviation of this sampling distribution

$$\mu = \frac{2 + 4 + 10 + 12 + 18}{5} = 9.2$$

$$\sigma^2 = \frac{(2 - 9.2)^2 + (4 - 9.2)^2 + (10 - 9.2)^2 + (12 - 9.2)^2 + (18 - 9.2)^2}{5} = 32.96$$

$$\sigma = 5.74$$

Combinations of the Samples	The mean of the samples ($\bar{X}$)
2,4,10	16/3
2,4,12	18/3
2,4,18	24/3
2,10,12	24/3
2,10,18	30/3
2,12,18	32/3
4,10,12	26/3
4,10,18	32/3
4,12,18	34/3
10,12,18	40/3

$$E(\bar{X}) = \frac{\frac{16}{3} + \frac{18}{3} + \dots + \frac{40}{3}}{10} = 9.2$$

$$\sigma_{\bar{x}} = \sqrt{\frac{(\frac{16}{3} - 9.2)^2 + (\frac{18}{3} - 9.2)^2 + \dots + (\frac{40}{3} - 9.2)^2}{10}} = 2.34$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{5.74}{\sqrt{3}} \sqrt{\frac{5-3}{5-1}} = 2.34$$

Theorem: If a population has a normal distribution with mean μ and standard deviation σ , then the random variable

$$z = \frac{\bar{X} - E[\bar{X}]}{\sigma_{\bar{x}}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has a standard normal distribution.

Example: The height of male students in a university is normally distributed with a variance of 19. If 25 male students are selected from this community, what is the probability that the difference between the sample mean and the population mean is least than 2 cm?

$$\begin{aligned} P[|\bar{X} - \mu| \leq 2] &= P\left[\frac{|\bar{X} - \mu|}{\sigma/\sqrt{n}} \leq \frac{2}{\sigma/\sqrt{n}}\right] = P\left[\frac{|\bar{X} - \mu|}{\sigma_{\bar{X}}} \leq \frac{\sqrt{n} \cdot 2}{\sigma}\right] = P\left[|Z| \leq \frac{\sqrt{25} \cdot 2}{\sqrt{19}}\right] \\ &= P[|Z| \leq 2.29] = P[-2.29 \leq Z \leq 2.29] = 2(0.4890) = 0.9780 \end{aligned}$$

Theorem: If a population has a normal distribution with mean μ (its standard deviation σ is unknown), then the random variable

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a **t** distribution with degree of freedom **$n-1$** . Here, **S** is the standard deviation of the sample and defined by

$$S = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n-1}}$$

t (Student's t) Distribution

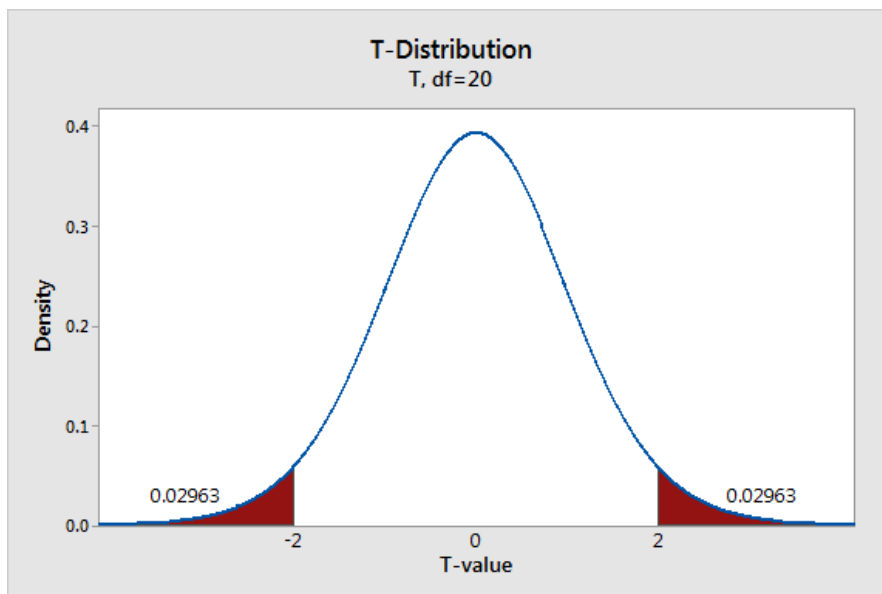
The probability density function of t distribution is given by

$$f(t; v) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi}\Gamma(v/2)} \left(1 + \frac{t^2}{v}\right)^{-\left(\frac{v+1}{2}\right)}, -\infty < t < \infty$$

where positive integer v is called degree of freedom. The mean and variance of the random variable t is:

$$E(t) = 0, \quad \text{Var}(t) = \frac{v}{v-2},$$

The t distribution is symmetric and is denoted by t_v for the degree of freedom v . When v increases, the t distribution approximates the standard normal distribution.



Example: A random sample of size 16 is selected from a population that is normally distributed. If the mean and the standard deviation of the sample is 75 and 8.5, respectively, then find the probability that the mean of the population is more than 78.5.

$$P(\mu > 78.5) = P(-\mu < -78.5) = P\left(\frac{\bar{X} - \mu}{S/\sqrt{n}} < \frac{\bar{X} - 78.5}{S/\sqrt{n}}\right) = P\left(t_{15} < \frac{75 - 78.5}{8.5/\sqrt{16}}\right) \\ = P(t_{15} < -1.647)$$

Since the area corresponding to the value 1.647 with the degree of freedom 15 does not exist in the **t** table, we will use the interpolation for:

$$P(t_{15} < -1.341) = 0.10$$

$$P(t_{15} < -1.753) = 0.05$$

1.341 0.1

1.647 Y

1.753 0.05

$$\frac{1.753 - 1.341}{1.647 - 1.341} = \frac{0.05 - 0.10}{Y - 0.10}, \quad Y = 0.0629$$

$$\rightarrow P(\mu > 78.5) = 0.0629$$

The populations are assumed to be normally distributed in the theorems given above. If the population is not normally distributed (or its distribution is unknown), we can apply the Central Limit Theorem which is given below.

The Central Limit Theorem: If a population has **any (unknown)** distribution with mean μ and standard deviation σ , then sample mean $\bar{X}$ has normal distribution for the limiting form of the sample size as $n \rightarrow \infty$. Then, the random variable

$$z = \frac{\bar{X} - E[\bar{X}]}{\sigma_{\bar{x}}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has the standard normal distribution when $n \rightarrow \infty$.

In practice, if $n \geq 30$ and the distribution of the population is unknown, then the random variable

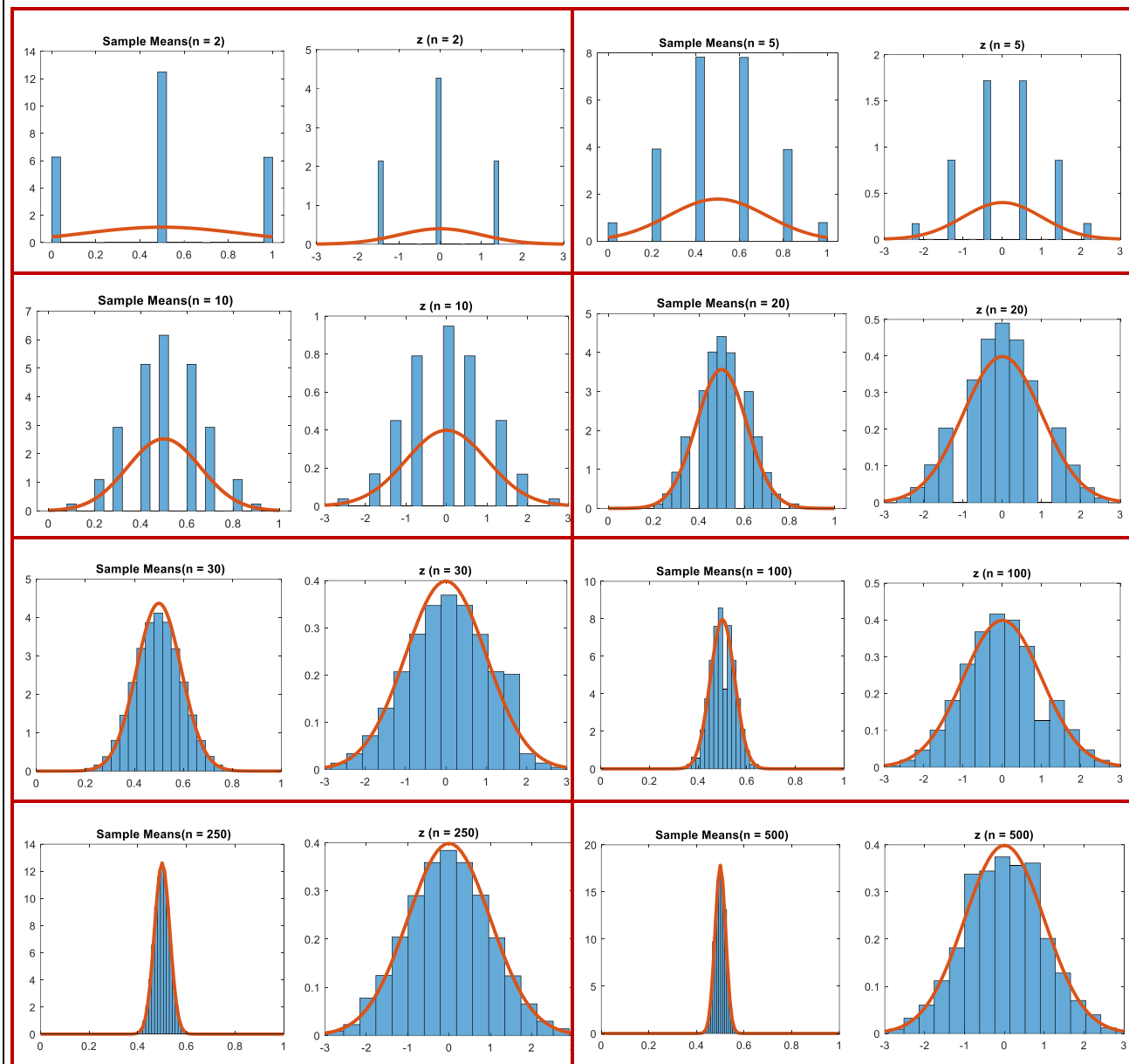
$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

is approximately distributed standard normally. If we don't know the variance of the population, we can assume that the random variable

$$z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

is distributed standard normally. Note that the last implication is not based on a theoretical derivation; it is just a practical approximation.

Simulation: Suppose that a coin is tossed 1 million times ($N = 1,000,000$). Tail and Head are represented by 0 and 1, respectively. When n was chosen as 2, 5, 10, 20, 30, 100, 250 and 500, the simulation produced the following histograms for sample means and their transformations to standard normal distribution. As seen, it approximates normal distribution for a large value of n . (The mean and standard deviation of the population is 0.5.)



Example: The bottles are filled on average 75 cl soda by a machine in a factory. It is known that the standard deviation of the amount of soda in the bottles is 4.8. If a random sample of size 42 units is selected, find the probability that the sample mean is more than 77 cl.

$$P(\bar{X} > 77) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{77 - \mu}{\sigma/\sqrt{n}}\right) = P\left(Z > \frac{77 - 75}{4.8/\sqrt{42}}\right) = P(Z > 2.7) = 0.5 - 0.4965 = 0.0035$$

2. Sampling Distribution of the Sample Variance

If all possible samples of size n are selected into a population of size N and the variance of each sample is calculated, we obtain **the sampling distribution of the sample variances**.

Theorem: If S^2 is the variance of a random sample of size n that is selected from a population which has normal distribution with variance σ^2 , the random variable

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

has the Chi-Squared distribution with the degree of freedom $n-1$.

Chi-Squared Distribution

The probability density function of χ^2 distribution is given by

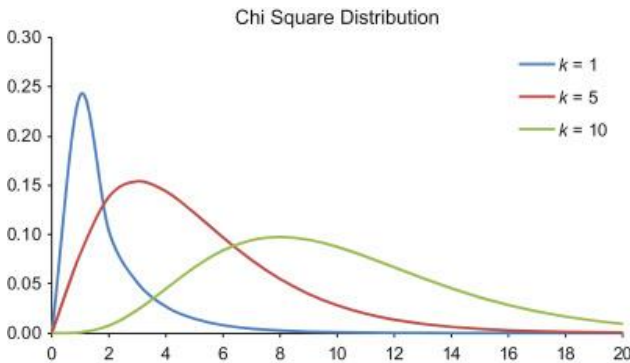
$$f(x; v) = \frac{x^{v/2-1} e^{-x/2}}{\Gamma(v/2) 2^{v/2}}, 0 \leq x < \infty$$

where positive integer v is called degree of freedom. The mean and variance of the random variable χ^2 is:

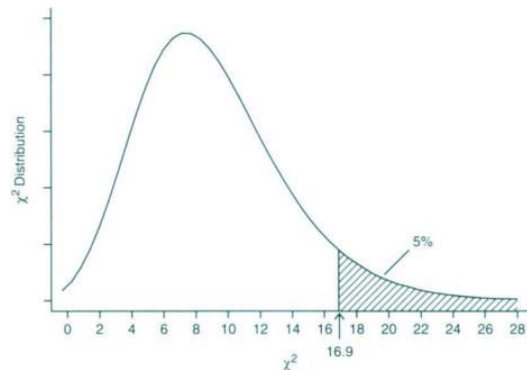
$$E(x) = v,$$

$$\text{Var}(t) = 2v$$

χ^2 distribution is not symmetric and is denoted by χ_v^2 for the degree of freedom v . When v increases, χ^2 distribution approximates a normal distribution.



(a)



(b)

a) Chi-squared distributions for $v = 1, 5, 10$. b) Chi-squared distributions $v = 9$
<https://www.sciencedirect.com/topics/mathematics/chi-square-distribution>

Example: A random sample of size 16 is selected from a population that is normally distributed. If the variance of the sample is 27.6, then find the probability that the variance of the population is more than 29.

$$\begin{aligned} P(\sigma^2 > 29) &= P\left(\frac{1}{\sigma^2} < \frac{1}{29}\right) = P\left(\frac{(n-1)S^2}{\sigma^2} < \frac{(n-1)S^2}{29}\right) = P\left(\chi_{15}^2 < \frac{(n-1)S^2}{29}\right) \\ &= P\left(\chi_{15}^2 < \frac{(16-1)27.6}{29}\right) = P(\chi_{15}^2 < 14.3) = 1 - 0.50 = 0.50 \end{aligned}$$

3. Sampling Distribution of the Sample Proportion

We know that the binomial distribution can be used for the selection with replacing them back from a population or for the selection without replacing them back from a large population. If X is a random variable for the binomial distribution, then the mean and the variance of X is

$$E(X) = np,$$

$$\text{Var}(X) = npq,$$

where p is the probability of success for a Bernoulli trial, and it is called **the population proportion**.

Since X represents the number of successes in the Binomial distribution, if we select a sample of size n from a population and the number of the members having the requested characteristics in the sample is X , then the proportion

$$\hat{p} = \frac{X}{n}$$

is called **the success proportion of the sample**. In this case, the sample proportion $\hat{p}$ is a sample statistic while the population proportion p is a population parameter.

The mean and the variance of the sample proportion is:

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n}E(X) = \frac{1}{n}np = p,$$

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{X}{n}\right) = \frac{1}{n^2}\text{Var}(X) = \frac{1}{n^2}npq = \frac{pq}{n},$$

and its standard deviation is

$$\sigma_{\hat{p}} = \sqrt{\frac{pq}{n}}.$$

Theorem: According to the Central Limit Theorem, $\hat{p}$ has normal distribution distribution for the limiting form of the sample size as $n \rightarrow \infty$ since $\hat{p} = \frac{X}{n} = \frac{(Y_1 + Y_2 + \dots + Y_n)}{n}$ where Y 's are random variable of Bernoulli distribution. Then, the random variable

$$z = \frac{\hat{p} - E(\hat{p})}{\sigma_{\hat{p}}} = \frac{\hat{p} - p}{\sqrt{pq/n}}$$

has the standard normal distribution when $n \rightarrow \infty$.

Example: If a balanced die is rolled 1500 times, find the probability that the difference between the frequency proportion of occurring 5 and the probability of occurring 5 is less than 0.02.

$$\begin{aligned} P(|\hat{p} - p| < 0.02) &= P\left(\frac{|\hat{p} - p|}{\sqrt{pq/n}} < \frac{0.02}{\sqrt{pq/n}}\right) = P\left(|z| < \frac{0.02\sqrt{1500}}{\sqrt{\frac{1}{6} \cdot \frac{5}{6}}}\right) = P(|z| < 2.08) \\ &= P(-2.08 < z < 2.08) = 2 * 0.4812 = 0.9624 \end{aligned}$$

Note that, when we seek the population proportion p (also q) in practice, we can use $\hat{p}$ and $\hat{q}$ instead of p and q in the above theorem.